

Seeing Like an Organization -- Follow-Up Study (#286030)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 3 authors.

Pre-registered on:

2026/04/19 00:55 (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

This study ("Study 2") replicates and extends Study 1 (AsPredicted# 269,965), which examined whether shared cognitive schemas can both enable and hinder adaptation under environmental change. Participants are randomized into small groups in a 2×2 design: Schema vs. Non-Schema $\times$ type of round-4 environmental change. As in Study 1, rounds 1-2 are condition-specific: in the Schema condition, they share a consistent feature structure that encourages schema formation; in the Non-Schema condition, they lack such structure and discourage schema formation. Round 3 introduces an environmental change that remains solvable using schemas developed in rounds 1-2 of the Schema condition. Round 4 varies environmental change type: either a change that appears compatible with the Schema-condition schema but is not ("schema-non-obvious," identical to Study 1), or a change that is clearly incompatible with that schema ("schema-obvious").

Hypothesis 1. Schema-condition groups are more likely to use a schema in round 4 under schema-non-obvious than under schema-obvious change.

3) Describe the key dependent variable(s) specifying how they will be measured.

Each round of the coordination task consists of two stages. In the Discussion Stage, participants discuss a set of 8 images and assign each image a unique name. In the Recall Stage, participants individually recall the names assigned to each image. Outcomes will be analyzed at the group level. For each round, we compute: (1) Accuracy: the number of images (0-8) for which all group members recall the same unique name in the Recall Stage (i.e., all members provide the same label for that image, and that label is not repeated across images); (2) Schema use: the number of images (0-8) for which all group members use a schema-based name in the Recall Stage. A schema-based name is a label generated by applying a consistent feature-to-name mapping across images. Three independent coders will determine for each image whether the recalled label is schema-based. Disagreements will be resolved by a majority vote. For analyses that require a binary classification, a group will be coded as using a schema in a given round if all group members used a schema-based name for more than half of the images.

4) How many and which conditions will participants be assigned to?

We will randomly assign participants approximately evenly to one of 4 conditions in our 2×2 design: Schema vs. Non-Schema and Schema-non-obvious vs. Schema-obvious change.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

To test Hypothesis 1, we will compare the proportion of groups that use a schema in round 4 under Schema-non-obvious vs. Schema-obvious change using a one-tailed Fisher's exact test at $\alpha = 0.05$.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will exclude groups where: (1) One or more participants are reported as not participating in the Discussion Stage, and don't check in within 60 seconds, or are reported as not participating more than once; (2) A group will be excluded if either participant leaves more than two recall responses blank in any single round; or, (3) Video recordings or text entry results reveal cheating. Examples: recording names and images during the Discussion Stage of the round, communicating during the Recall Stage of the round, or other behaviors prohibited by the study instructions.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Our final sample size for collected data will be approximately $N = 256$ participants after excluding observations listed in item 6 above (approximately 64 per condition).

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

(1) Pilot data were collected prior to pre-registration for QA purposes. It will not be used in the analysis. (2) At the end of the experiment, participants will be asked to complete a demographic survey including questions on Age, Gender, Education, and Race. These data may be used for ex post exploratory analysis. They will also be asked to answer open-ended questions explaining their choices in stage 2. This data may be used in future exploratory analysis. (3) Participants will be able to communicate with one another via video chat and text boxes during the first stage. This video and textual data, coupled with the names assigned by participants to images in the first stage, will also be collected for potential future exploratory analysis.